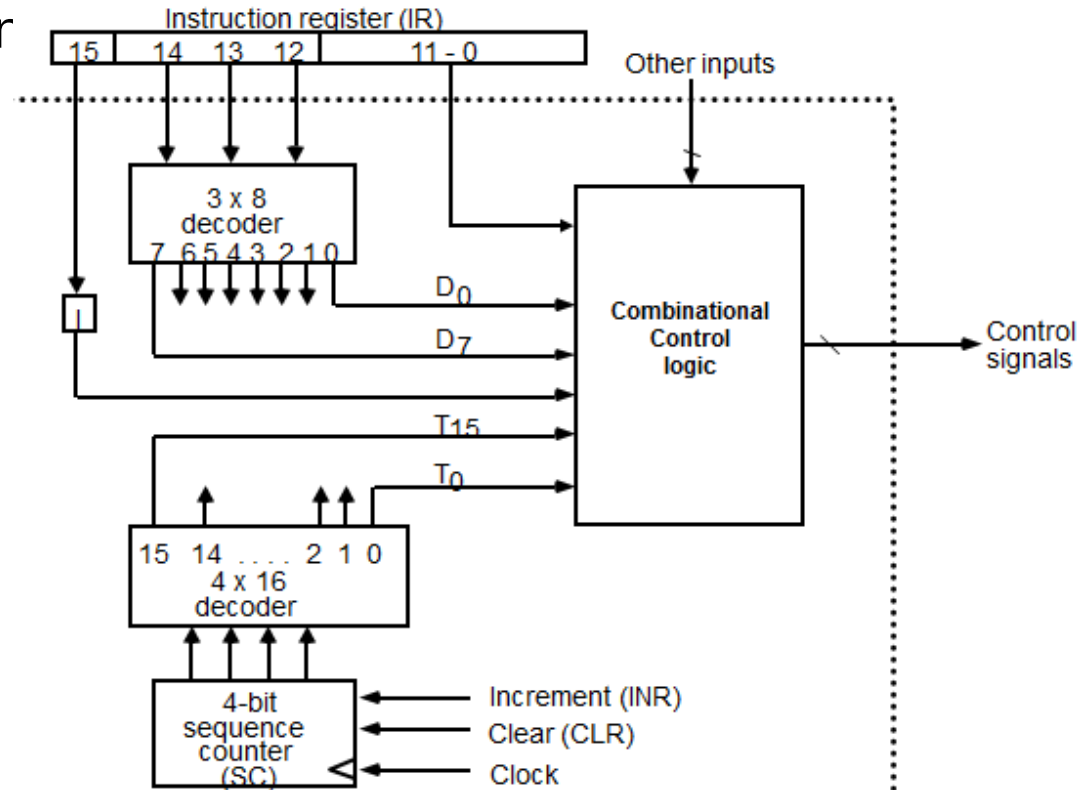


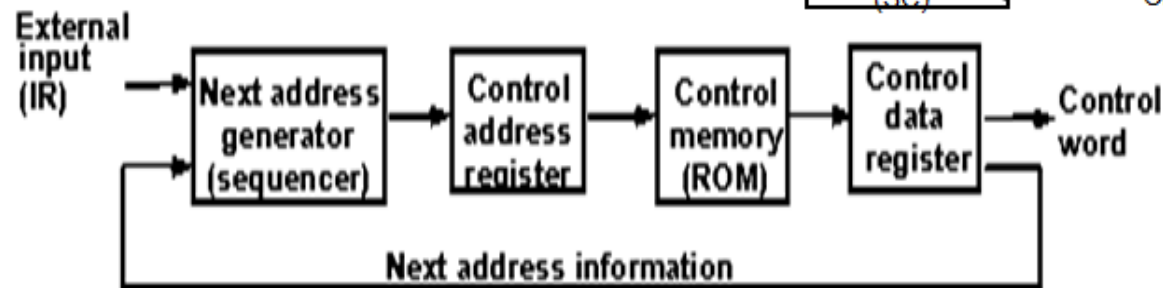
COMPARISON OF CONTROL UNIT IMPLEMENTATIONS

Control Unit Implementations

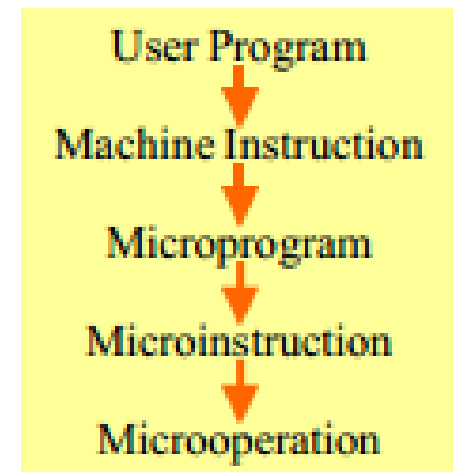
- Hard-wired CU



- Micro Programmed CU



- Routine (Microprogram)
 - Microinstruction are stored in control memory *in groups*.
- Subroutine
 - *program used by other ROUTINES*
- Control Address Register
 - Specify the address of the microinstruction
- Control Data Register (= *Pipeline Register*)
 - Hold the microinstruction read from control memory.
 - Allows the execution of the microoperations specified by the control word *simultaneously* with the generation of the next microinstruction

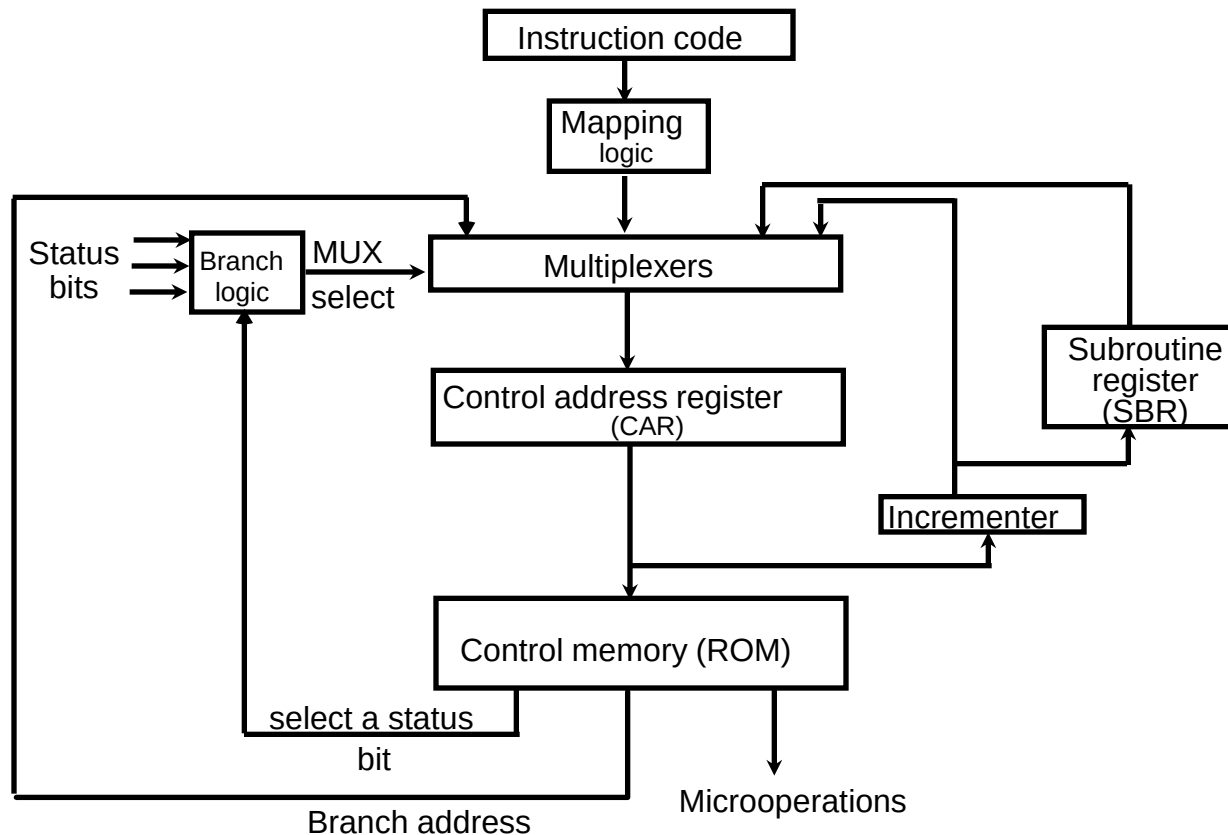


Address Generation in Control Memory

- *Sequencer (Microprogram Sequencer)*
 - It determines the address of the microinstruction in the control memory to be executed in the next clock cycle.
 - Next address of the next microinstruction can be specified in several ways depending on the sequencer input.
- In-line Sequencing
- Branch
- Conditional Branch
- Subroutine
- Loop
- Instruction OP-code mapping



Address Sequencing



Sequencing Capabilities Required in a Control Storage

- Incrementing of the control address register
- Unconditional and conditional branches
- A mapping process from the bits of the machine instruction to an address for control memory
- A facility for subroutine call and return



Address Sequencing

- CAR : Control Address Register.
 - CAR receive the address from 4 different paths:
 - 1) Incrementer
 - 2) Branch address from control memory
 - 3) Mapping Logic
 - 4) SBR : Subroutine Register.
- SBR : Subroutine Register.
 - Return Address can not be stored in ROM
 - Return Address for a subroutine is stored in SBR (a special Register).
- Conditional Branching.
 - Status Bits.
 - Control the conditional branch decisions generated in the *Branch Logic*.
 - Branch Logic.
 - Test the specified condition and Branch to the indicated address if the condition is met ; otherwise, the control address register is just incremented.
 - Status Bit Test.
 - Multiplexer

